
E-Commerce Sales Dashboard

Analysis

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Business Analysis Report

1. Descriptive Analysis (What Happened?)

Descriptive analysis explains the current performance of the business based on the available data.

Overall Business Performance

- **Total Sales Revenue:** \$2,174,394.70
- **Total Orders:** 2,000
- **Average Order Value:** \$1,087.20
- **Median Order Value:** \$868.72
- **Average Quantity Purchased:** 2.48 items
- **Discount Offered:** Average 15% (0%–30%)

The business has generated over **\$2.17 million** in sales with a healthy average order value, indicating strong customer spending.

Customer Analysis

Gender-wise Analysis

- Female customers contribute **56.63%** of total sales.
- Male customers contribute **43.37%**.
- Female customers also place more orders (54.8%).

Insight

Female customers are currently the primary revenue drivers and represent the strongest customer segment.

Age-wise Analysis

Age Group	Observation
18-25	Lowest sales contribution
26-35	Steady growth
36-45	High revenue
46-55	Highest revenue
56-65	Slight decline

The **46-55** age group generates the highest sales, followed by **36-45**.

Insight

Customers between **36-55 years** are the company's most valuable customers.

Regional Analysis

Region Contribution

North	29.71%
South	29.21%
East	28.62%
West	12.45%

Insight

North, South, and East together contribute nearly **88%** of total revenue, while West significantly underperforms.

Category Analysis

Category Contribution

Sports	24.93%
Electronics	23.19%
Home	19.65%
Clothing	17.44%
Beauty	14.79%

Top Subcategories

Sports

- Fitness
- Accessories
- Team Sports

Electronics

- Computer
- Accessories
- Audio

Home

- Decor
- Kitchen

Clothing

- Outerwear
- Footwear

Beauty

- Haircare
- Skincare

Insight

Sports and Electronics together contribute nearly **48%** of total business revenue, making them the company's strongest categories.

Beauty contributes the least among all categories.

Loyalty Analysis

Level	Contribution
Platinum	30.98%
Bronze	24.04%
Gold	23.81%
Silver	21.18%

Insight

Platinum customers generate almost one-third of total revenue, highlighting the importance of customer retention.

Top Products Analysis

Top-selling products include:

- Portable Power Bank
- Gaming Mechanical Keyboard
- Wireless Mouse
- Basketball
- Men's Casual Sneakers

Most top-selling products belong to **Electronics** and **Sports** categories.

Sales Trend Analysis

Highest sales months:

- March 2025
- March 2024
- October 2024
- November 2024

Lowest sales:

- October 2023
- October 2025

Insight

The business experiences seasonal fluctuations with several months crossing **\$100,000** in sales.

Discount Analysis

- Discounts range between **0% and 30%**
- Average discount is **15%**

Observation

Higher discounts do **not necessarily** generate higher sales.

Some products perform exceptionally well with little or no discount, indicating that product demand and category strength influence sales more than discounts alone.

Statistical Analysis

- Mean sales value: **\$1,087**
 - Standard deviation indicates high variability in order values.
 - Sales distribution is positively skewed because a small number of high-value orders increase the average.
 - Customers typically purchase **2 items per transaction**.
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2. Diagnostic Analysis (Why Did It Happen?)

Why is Sports the highest-selling category?

- Increasing interest in fitness activities
 - High demand for accessories
 - Repeat purchases
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Why is Electronics performing well?

- Higher unit prices
 - Strong demand for computers and accessories
 - Technology products naturally generate higher revenue
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Why is Beauty the lowest contributor?

Possible reasons:

- Lower-priced products
 - High market competition
 - Limited demand in some subcategories
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Why are Female customers generating more revenue?

- Higher order volume
 - Slightly higher spending per transaction
 - Better engagement with available product categories
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Why does the 36-55 age group dominate sales?

This age group generally has:

- Stable income
 - Higher purchasing power
 - Family spending responsibilities
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Why is West region underperforming?

Possible reasons:

- Lower market penetration
 - Fewer promotional activities
 - Product assortment mismatch
 - Distribution challenges
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Why are Platinum members contributing the most?

Because they are:

- Repeat customers
- High-value shoppers

- More engaged with the platform
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Why don't higher discounts always increase sales?

The analysis suggests that:

- Brand value
- Product quality
- Customer demand

have a stronger impact on purchasing decisions than discount percentage alone.

3. Predictive Analysis (What Will Likely Happen?)

If current trends continue:

Category Prediction

- Sports will likely remain the top category.
 - Electronics will continue strong growth.
 - Beauty may remain the lowest contributor unless strategies change.
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Customer Prediction

The 36-55 age group will continue generating maximum revenue.

Female customers are likely to remain the dominant customer segment.

Regional Prediction

North, South, and East will continue driving sales.

Without intervention, West may continue to lag behind.

Loyalty Prediction

Platinum customers will continue contributing the highest revenue.

Improving loyalty programs will increase repeat purchases and customer lifetime value.

Discount Prediction

Increasing discounts alone is unlikely to significantly increase revenue and may reduce profit margins.

Sales Prediction

Seasonal peaks are likely to continue around months that historically performed well, such as March and October–November.

4. Prescriptive Analysis (What Should the Business Do?)

Recommendation 1: Focus on High-Performing Categories

Increase investment in:

- Sports
- Electronics

through:

- More inventory
- Premium products
- Better marketing

Expected Impact: Higher revenue growth.

Recommendation 2: Improve Beauty Category Performance

- Product bundles
- Influencer marketing
- Seasonal campaigns
- Subscription offers

Expected Impact: Increased category contribution.

Recommendation 3: Expand in the West Region

- Localized advertising
- Region-specific offers
- Better logistics

Expected Impact: Balanced regional growth.

Recommendation 4: Strengthen Loyalty Programs

- Exclusive Platinum benefits
- Upgrade incentives for Bronze and Silver
- Personalized rewards

Expected Impact: Higher customer retention and repeat purchases.

Recommendation 5: Increase Average Order Value

Introduce:

- Product bundles
- Cross-selling
- Upselling
- Free shipping thresholds
- Buy More, Save More offers

Expected Impact: Higher basket value and increased revenue.

Recommendation 6: Target the 36-55 Age Segment

Develop personalized campaigns featuring:

- Premium products
- Family-oriented bundles
- Lifestyle products

Expected Impact: Improved conversions and customer lifetime value.

Recommendation 7: Grow the 18-25 Customer Segment

- Student discounts
- Social media campaigns
- Affordable product lines

Expected Impact: Acquisition of long-term customers.

Recommendation 8: Optimize Discount Strategy

Instead of blanket discounts:

- Offer category-specific promotions
- Reward loyal customers
- Test discount effectiveness before scaling

Expected Impact: Better profitability while maintaining sales.

Recommendation 9: Leverage Seasonal Trends

Plan inventory and marketing around historically strong months such as:

- March
- October
- November

Expected Impact: Better stock utilization and higher seasonal revenue.

Recommendation 10: Use Data-Driven Personalization

Segment customers by:

- Gender
- Age
- Region
- Loyalty level
- Purchase history

Deliver personalized recommendations and offers.

Expected Impact: Higher conversion rates, increased repeat purchases, and improved marketing ROI.



Executive Summary

The e-commerce business generated **\$2.17 million** from **2,000 transactions**, demonstrating strong overall performance. **Sports (24.93%)** and **Electronics (23.19%)** are the primary revenue-driving categories, together contributing nearly **48% of total sales**. **Female customers**, **customers aged 36–55**, and **Platinum loyalty members** are the most valuable customer segments. Geographically, the **North, South, and East** regions perform strongly, while the **West** region offers the greatest opportunity for expansion. The analysis also shows that **discounts alone do not guarantee higher sales**; product demand, category strength, and customer loyalty have a greater influence on revenue. By focusing on high-performing categories, strengthening loyalty programs, expanding in underperforming regions, increasing average order value through cross-selling and bundling, and adopting personalized, data-driven marketing strategies, the business can improve both **sales growth and long-term profitability**.

Ecommerce Sales Analysis Dashboard

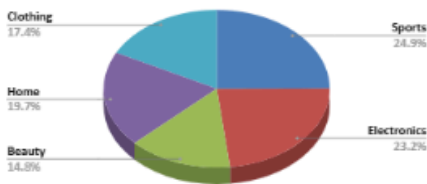
SUM of Total_Amount vs. Order_Date - Year-Month



Customer Age Range Sales



Category Vs Sales



Gender Wise Sales



Total Sales
\$2,174,394.70

Total orders
2000

Top 15 Results					Category	Sub_Category	Store_Name	Product
No	Brand	M of Total_Amount	M of Total_Amount	Percentage				
1	umblings and Soi	\$41,852.50	1.92%		Electronics	Mobile	ez, Adkins and K	Portable Power Bank
2	French Ltd	\$38,281.02	1.76%		Sports	Team Sports	Benson-Elliott	Shin Guards - Adult
3	Hogan-Lee	\$35,489.62	1.63%		Clothing	Footwear	Rivera-Warner	Men's Casual Sneakers
4	Lucero LLC	\$34,699.83	1.60%		Electronics	Computer	Frederick-Wright	Gaming Mechanical Keyboard
5	Smith-Charles	\$33,993.66	1.56%		Clothing	Footwear	Holloway Ltd	Women's High Heels
6	Larson Ltd	\$33,566.90	1.54%		Electronics	Computer	san, Jones and Pe	Wireless Mouse
7	Nolan-Snow	\$31,070.46	1.43%		Sports	Team Sports	Wagner and Bal	Basketball
8	Cline-Mccarthy	\$30,236.79	1.39%		Beauty	Haircare	Gray Ltd	Men's Hair Styling Gel
9	Phillips and Sons	\$28,572.69	1.31%		Clothing	Outerwear	Gray Ltd	Women's Formal Blazer
10	l, Murphy and Wi	\$28,324.05	1.30%		Sports	Accessories	Guerrero-Schmitt	Soccer Training Cone Set
11	Thomas LLC	\$28,285.61	1.30%		Electronics	Appliances	Martinez-Mullins	4K Home Projector
12	Taylor-Elliott	\$28,237.93	1.30%		Beauty	Makeup	Benson-Elliott	Face Foundation Kit
13	toffman-William	\$28,094.22	1.29%		Beauty	Skincare	Johnson-Wright	Vitamin E Body Lotion
14	Hernandez Inc	\$27,749.20	1.28%		Home	Decor	Frederick-Wright	Essential Oil Diffuser
15	Soto-Ellis	\$27,622.96	1.27%		Sports	Accessories	san, Jones and Pe	Outdoor Sports Water Bottle

Loyalty_Level All ▾

Region All ▾

Product_SubCategory All ▾

Year All ▾

Payment_Type All ▾